JHON.py - C:/Users/CSSC/Downloads/JHON.py (3.13.10)

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```
import tkinter as tk
from tkinter import ttk, messagebox
import sqlite3

# ========================
# DATABASE
# ========================
conn = sqlite3.connect("students.db")
cursor = conn.cursor()

cursor.execute("""
CREATE TABLE IF NOT EXISTS students (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    name TEXT NOT NULL,
    age INTEGER NOT NULL,
    course TEXT NOT NULL
)
""")

conn.commit()


# ========================
# FUNCTIONS
# ========================

# CREATE
def add_student():
    name = name_entry.get().strip()
    age = age_entry.get().strip()
    course = course_entry.get().strip()

    if name == "" or age == "" or course == "":
        messagebox.showwarning(
            "Warning",
            "Please fill in all fields."
        )
        return

    try:
        age = int(age)
    except ValueError:
        messagebox.showerror(
            "Error",
            "Age must be a number."
        )
        return
```

```
y - C:/Users/CSSC/Downloads/JHON.py (3.13.10)*
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)
return

or.execute(
"INSERT INTO students (name, age, course) VALUES (?, ?, ?)",
(name, age, course)


.commit()

agebox.showinfo(
"Success",
"Student added successfully."


r_fields()
lay_students()



lay_students():
item in tree.get_children():
tree.delete(item)

or.execute("SELECT * FROM students")
lents = cursor.fetchall()

index, student in enumerate(students):
tag = "evenrow" if index % 2 == 0 else "oddrow"

tree.insert(
    "",
    tk.END,
    values=student,
    tags=(tag,)
)


:
te_student():
cted = tree.selection()

ot selected:
messagebox.showwarning(
    "Warning",
    "Please select a student to update."
)
return
```

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```
# ==========================
# INPUT FRAME
# ==========================

input_frame = tk.Frame(
    root,
    bg="white",
    bd=2,
    relief="groove"
)

input_frame.pack(
    pady=5,
    padx=20,
    fill="x"
)


# ==========================
# NAME
# ==========================

tk.Label(
    input_frame,
    text="Name:",
    font=("Arial", 11, "bold"),
    bg="white",
    fg=LABEL_COLOR
).grid(
    row=0,
    column=0,
    padx=10,
    pady=8,
    sticky="e"
)

name_entry = tk.Entry(
    input_frame,
    width=35,
    font=("Arial", 11),
    bg=ENTRY_BG,
    fg="#212121",
    relief="solid",
    bd=1
)

name_entry.grid(
    row=0,
    column=1
```

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```
name_entry.grid(
    row=0,
    column=1,
    padx=10,
    pady=8
)


# ==========================
# AGE
# ==========================

tk.Label(
    input_frame,
    text="Age:",
    font=("Arial", 11, "bold"),
    bg="white",
    fg=LABEL_COLOR
).grid(
    row=1,
    column=0,
    padx=10,
    pady=8,
    sticky="e"
)

age_entry = tk.Entry(
    input_frame,
    width=35,
    font=("Arial", 11),
    bg=ENTRY_BG,
    fg="#212121",
    relief="solid",
    bd=1
)

age_entry.grid(
    row=1,
    column=1,
    padx=10,
    pady=8
)


# ==========================
# COURSE
# ==========================
```

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```
# ==========================
# COURSE
# ==========================

tk.Label(
    input_frame,
    text="Course:",
    font=("Arial", 11, "bold"),
    bg="white",
    fg=LABEL_COLOR
).grid(
    row=2,
    column=0,
    padx=10,
    pady=8,
    sticky="e"
)

course_entry = tk.Entry(
    input_frame,
    width=35,
    font=("Arial", 11),
    bg=ENTRY_BG,
    fg="#212121",
    relief="solid",
    bd=1
)

course_entry.grid(
    row=2,
    column=1,
    padx=10,
    pady=8
)


# ==========================
# BUTTON FRAME
# ==========================

button_frame = tk.Frame(
    root,
    bg=BG_COLOR
)

button_frame.pack(pady=15)


# ==========================
```

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```
# ==========================
# BUTTON FRAME
# ==========================

button_frame = tk.Frame(
    root,
    bg=BG_COLOR
)

button_frame.pack(pady=15)


# ==========================
# ADD BUTTON
# ==========================

tk.Button(
    button_frame,
    text="Add",
    width=12,
    font=("Arial", 10, "bold"),
    bg=ADD_COLOR,
    fg=BUTTON_TEXT,
    activebackground="#1E8449",
    activeforeground="white",
    relief="flat",
    cursor="hand2",
    command=add_student
).grid(
    row=0,
    column=0,
    padx=5
)


# ==========================
# UPDATE BUTTON
# ==========================

tk.Button(
    button_frame,
    text="Update",
    width=12,
    font=("Arial", 10, "bold"),
    bg=UPDATE_COLOR,
    fg=BUTTON_TEXT,
    activebackground="#D68910",
```

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```
# ==========================
# UPDATE BUTTON
# ==========================

tk.Button(
    button_frame,
    text="Update",
    width=12,
    font=("Arial", 10, "bold"),
    bg=UPDATE_COLOR,
    fg=BUTTON_TEXT,
    activebackground="#D68910",
    activeforeground="white",
    relief="flat",
    cursor="hand2",
    command=update_student
).grid(
    row=0,
    column=1,
    padx=5
)


# ==========================
# DELETE BUTTON
# ==========================

tk.Button(
    button_frame,
    text="Delete",
    width=12,
    font=("Arial", 10, "bold"),
    bg=DELETE_COLOR,
    fg=BUTTON_TEXT,
    activebackground="#C0392B",
    activeforeground="white",
    relief="flat",
    cursor="hand2",
    command=delete_student
).grid(
    row=0,
    column=2,
    padx=5
)


# ==========================
# CLEAR BUTTON
```

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```
)


# ============================
# CLEAR BUTTON
# ============================

tk.Button(
    button_frame,
    text="Clear",
    width=12,
    font=("Arial", 10, "bold"),
    bg=CLEAR_COLOR,
    fg=BUTTON_TEXT,
    activebackground="#626567",
    activeforeground="white",
    relief="flat",
    cursor="hand2",
    command=clear_fields
).grid(
    row=0,
    column=3,
    padx=5
)


# ============================
# TREEVIEW STYLE
# ============================

style = ttk.Style()
style.theme_use("clam")


# TABLE HEADING
style.configure(
    "Treeview.Heading",
    background="#1F618D",
    foreground="white",
    font=("Arial", 10, "bold"),
    padding=8
)


# TABLE BODY
style.configure(
    "Treeview",
    background="white",
    foreground="#212121",
```

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```
style.configure(
    "Treeview.Heading",
    background="#1F618D",
    foreground="white",
    font=("Arial", 10, "bold"),
    padding=8
)


# TABLE BODY
style.configure(
    "Treeview",
    background="white",
    foreground="#212121",
    rowheight=30,
    fieldbackground="white",
    font=("Arial", 10)
)


# SELECTED ROW
style.map(
    "Treeview",
    background=[
        ("selected", "#5DADE2")
    ],
    foreground=[
        ("selected", "white")
    ]
)


# ==========================
# TABLE
# ==========================

tree = ttk.Treeview(
    root,
    columns=("ID", "Name", "Age", "Course"),
    show="headings"
)

tree.heading(
    "ID",
    text="ID"
)

tree.heading(
    "Name",
```

```
    columns=("ID", "Name", "Age", "Course"),
    show="headings"
)

tree.heading(
    "ID",
    text="ID"
)

tree.heading(
    "Name",
    text="Name"
)

tree.heading(
    "Age",
    text="Age"
)

tree.heading(
    "Course",
    text="Course"
)


# COLUMN WIDTHS
tree.column(
    "ID",
    width=60,
    anchor="center"
)

tree.column(
    "Name",
    width=200
)

tree.column(
    "Age",
    width=80,
    anchor="center"
)

tree.column(
    "Course",
    width=220
)
```

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```
# ==========================
# COLORS
# ==========================

tree.tag_configure(
    "evenrow",
    background="#F4F9FC"
)

tree.tag_configure(
    "oddrow",
    background="#D6EAF8"
)


tree.pack(
    fill="both",
    expand=True,
    padx=20,
    pady=10
)


# ==========================
# SELECT EVENT
# ==========================

tree.bind(
    "<<TreeviewSelect>>",
    select_student
)


# ==========================
# DISPLAY EXISTING RECORDS
# ==========================

display_students()


# ==========================
# START APPLICATION
# ==========================

root.mainloop()


# ==========================
```

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```
tree.tag_configure(
    "evenrow",
    background="#F4F9FC"
)

tree.tag_configure(
    "oddrow",
    background="#D6EAF8"
)


tree.pack(
    fill="both",
    expand=True,
    padx=20,
    pady=10
)


# ========================
# SELECT EVENT
# ========================

tree.bind(
    "<<TreeviewSelect>>",
    select_student
)


# ========================
# DISPLAY EXISTING RECORDS
# ========================

display_students()


# ========================
# START APPLICATION
# ========================

root.mainloop()


# ========================
# CLOSE DATABASE
# ========================
```

IDLE Shell - C:/Users/CSSC/Downloads/CRIS.py (3.13.10)

File

>>>

>>>

)] on win32

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Student Management System

Student Management System

Name:

Age:

Course:

Add | Update | Delete | Clear

ID	Name	Age	Course

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File

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# ==
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# ==
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curs
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)
"""

conn


# ==
# FU
# ==

# CR
def
```

Student Management System

Student Management System

Name:

Age:

Course:

Add | Update | Delete | Clear

ID	Name	Age	Course
1	Chris Jhon Del Abejan	19	BSIT 2-B